WHAT SOFTWARE-WORK TELEMETRY IDENTIFIES: ATTAINABLE LABOUR BOUNDS AND DESIGN-BASED ESTIMATION

Abstract. Software-work telemetry records events whose relationship to human labour can change under AI assistance. We determine which labour totals are compatible with specified observations, keeping elapsed time to acceptance and retained output as separate endpoints. Interval totals or justified bounds, together with per-person capacity, constrain labour at a requested cutoff. Partitioning time at observation endpoints and target cutoffs reduces the continuous allocation problem exactly to a finite linear programme, including overlapping observations and shared capacity. Its extrema delimit all attainable totals under those constraints. Summing separate interval bounds can give a wider range. When labour is probability-sampled, fixed telemetry predictions can assist estimation through a design-unbiased residual correction. The fixed predictions can increase or reduce design variance, depending on their residuals. The labour bounds and sampling estimator require no probabilistic model of task execution. We supply analytical proofs and exact finite computational checks. No organisational data are analysed. The proposed measurements still require empirical validation.

1. What the observations determine

Suppose a team records three person-hours of work during a four-hour interval. How much occurred during its first two hours? If one person supplied the work, the answer lies between one and two person-hours: at most two fit after the cutoff, and at most two fit before it. Both limits are attainable. The interval total therefore identifies a range without identifying the allocation within it. Commit times alone may supply even less information, because the same event sequence can arise from different amounts of human activity.

The engineering question is how much labour the observations determine, and which additional measurements would narrow the answer. We address it for AI-assisted software work, defining all-role person-hours separately from elapsed time to first acceptance and reference output retained at follow-up. The accounting includes unsuccessful work and post-acceptance repair, while limiting each person's concurrent allocations to their available time. Our principal result represents all feasible labour totals by a finite linear programme whose constraints encode interval totals or justified bounds and shared per-person capacity. Minimising and maximising the desired total gives its identified range: every value between the extrema is attainable by an admissible allocation. Narrowing that range therefore requires additional observations or justified restrictions on those allocations.

When exact allocation totals are unavailable, a probability sample with known inclusion probabilities supports design-unbiased estimation, provided sampled allocations are measured accurately. Telemetry predictions fixed before sampling can assist through a weighted residual correction. They improve precision when the weighted residual total has lower design variance than the corresponding unassisted labour estimator. Telemetry can therefore assist estimation even when its events leave labour unidentified. The resulting sampling uncertainty concerns estimation of labour, whereas an identification interval describes the totals compatible with the specified observations. Estimating an intervention effect additionally requires a causal study design.

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Wright and Ziegler's Standard Coder approach [13] motivates the observation problem. It infers coding activity from commit timing and predicts the resulting labels from code-change features. Standard Coding Hours (SCH) can serve as a common reference index, but calibration to independently observed labour requires a separate argument. AI may alter both the event-to-activity relationship and the work represented by a code change. The SPACE framework places activity and efficiency within a broader account of developer productivity [7]. Mozannar et al. examine interaction time costs in AI-assisted programming [10].

Existing AI studies address different parts of this measurement problem. Peng et al. examine completion time among completers of a programming exercise [11]. Becker et al. study experienced maintainers in familiar repositories with a forecast-adjusted log-time contrast [2]. Cui et al. study weekly completed pull requests in organisational experiments [5], with assignment used as an instrument for adoption in their June 2025 manuscript [4, Section 4.1 and Table 3]. These differences in populations, interventions and outcomes prevent treating their estimates as interchangeable parameters. METR's subsequent account of selection and concurrent-agent measurement difficulties further motivates separate labour and elapsed-time records [3]. The calculations here use no data from these studies.

Sections 2 and 3 define the observation problem and endpoints. Section 4 derives the labour bounds, and Section 5 connects sampled labour to telemetry-assisted estimation. Section 6 sets out the measurement validation needed before an intervention comparison. The finite reduction and attainability arguments specialise standard optimisation and sampling results to the stated allocation constraints. Appendix A gives a shortest-path solution for one allocation stream.

2. Why event telemetry alone need not identify labour

Let a minute-indexed hidden variable $Z_t \in \{0,1\}$ indicate an activity state, and let $C_t \in \{0,1\}$ record a commit event. A fitted event model specifies both activity dynamics and the mechanism that produces commits from activity. Interpreting $Z_t = 1$ as "coding" requires validation against independently measured labour. Even before that validation, distinct activity laws can produce the same event law.

Proposition 2.1 (Failure of unrestricted identification)**.** *Suppose the states Z_t are independent Bernoulli variables with parameter $\pi \in (0,1)$, and, independently across minutes and of the states, B_t are Bernoulli variables with parameter $c \in (0,1)$. Define $C_t = Z_t B_t$. The complete law of the event process depends on (π, c) only through πc. In particular, $(\pi, c) = (1/2, 1/5)$ and $(1/4, 2/5)$ give identical observed laws but different expected active times.*

Proof. Each C_t is Bernoulli with parameter $\theta = \pi c$, and the C_t are independent. For every $n \geq 1$ and every binary string $(e_1, \ldots, e_n)$,

$$\mathbb{P}(C_1 = e_1, \ldots, C_n = e_n) = \theta^{\sum_t e_t}(1-\theta)^{n-\sum_t e_t}. \tag{1}$$

Both parameter pairs give $\theta = 1/10$. In contrast,

$$\mathbb{E}\left[\sum_{t=1}^{n} Z_t\right] = n\pi, \tag{2}$$

which is $n/2$ minutes in the first model and $n/4$ in the second. Thus the same observable distribution corresponds to two different target values. □

This construction belongs to the independent-state subfamily of a two-state hidden Markov model: both transition rows give the same next-state probability. It shows that the unrestricted family is not globally identifiable. Persistent or otherwise restricted models may still be identifiable. Restrictions on the activity law, informative code-change features or independent activity observations may distinguish cases that commit timing alone leaves

unresolved. An identification claim must state which restrictions or observations supply that information. Under AI assistance, commit production can continue while human attention is elsewhere, so the event-producing mechanism itself may change.

Agreement between model predictions leaves their calibration to independently observed labour untested. We next ask how interval observations constrain labour.

3. Task accounting with explicit boundaries

3.1. **The task, the assigned intervention and the person-hour.** Enrol task i at time s_i, when its requested behaviour, eligibility and acceptance criteria are recorded. Give the task an identity tied to its requirement, so commits and pull requests can change without changing the unit counted. Before observing outcomes, define how split tasks, merged tasks and scope changes will be handled. A parent requirement can retain its original identity while children record implementation activity. Counting both the parent and its children as delivered output would count the same work twice. Changes affecting the stated requirement or acceptance criteria require a versioned record and a stated analysis rule.

Before interventions are compared, the rule must fix task granularity, the unit receiving acceptance credit and the start of follow-up. Consider separately accepted components regrouped into a parent accepted only after its last component. Even with additive reference scores, this regrouping can change the output credited by a deadline. The comparison therefore depends on the task and acceptance rules as well as the score.

An *AI intervention* specifies available tools and model versions, eligible tasks, permitted autonomous actions, review responsibilities and release requirements. A comparison of interventions must specify what is assigned and record actual use separately, since difficult tasks may elicit more use and people may decline an assigned tool.

Let $a_{ijr}(t)$ be the fraction of person j's time allocated to task i in role r at calendar time t. With finite sets of tasks, people and roles, measurable allocation functions define labour at task age $u \geq 0$ by

$$a_{ijr}(t) \geq 0, \qquad \sum_{i,r} a_{ijr}(t) \leq 1, \qquad H_i(u) = \sum_{j,r} \int_{s_i}^{s_i+u} a_{ijr}(t)\, dt. \tag{3}$$

With time in hours, $H_i(u)$ has units of person-hours. Include every role contributing labour to the task, from specification and implementation through review, testing and repair. Prompting and integration also consume person-hours. Pair work can contribute two person-hours in one elapsed hour. For one person, two concurrent sessions cannot each receive a full hour of attention. State whether attended waiting belongs in the labour total. Unattended execution is recorded as agent resource use.

Equation (3) defines the labour target. Measuring it requires an attribution rule for shared review and platform work, with an overhead category for work that the observations cannot assign to particular tasks. Record agent instance-hours by resource class, tokens and charges separately from person-hours. Combining these resources requires an explicit economic conversion and changes the question from labour to cost.

Let T_i be elapsed time from enrolment to first acceptance. Starting this clock at enrolment includes queueing and any reasoning and specification between enrolment and the first commit. If the task is permanently abandoned before acceptance, put $T_i = \infty$. Abandonment is an outcome, whereas loss of observation is censoring. Record deployment separately whenever acceptance and deployment differ.

3.2. **A frozen reference and a follow-up assessment.** Assign a nonnegative reference score S_i^0 using a declared scoring-rule version and task representation. It may count tasks or functional units, or use frozen SCH weights. Retained output has the same unit as this score. Productivity divides that output by person-hours. The superscript 0 denotes the reference held fixed across interventions. A score expressed in person-hours needs calibration

to independently measured labour for a specified reference population, included roles, quality criterion and stopping rule. A predictor trained on imputed activity states targets those states, leaving its calibration to labour untested.

For a fixed-scope causal comparison, requirements recorded at intake are preferable to final code differences: AI can change the latter. A retrospective code score can still be reported, with its dependence on the implementation made explicit. A nonlinear scoring rule need not be additive, so splitting or squashing a change can alter its score. Score a canonical task representation once, or use an additive requirement rubric. A failed attempt, retry or revert must not generate another unit of retained output.

Choose an acceptance horizon $\tau > 0$ and a follow-up duration $\Delta \geq 0$. For accepted tasks, let $R_i(\Delta) \in [0, 1]$ be the assessed fraction of the accepted requirement still meeting the rubric at time $s_i + T_i + \Delta$. Retained reference output is

$$Q_i(\tau, \Delta) = \begin{cases} S_i^0 R_i(\Delta), & T_i \leq \tau, \\ 0, & T_i > \tau. \end{cases} \tag{4}$$

The rubric must distinguish an intentional replacement from a defect-driven loss and explain fractional scores. Surviving lines, successful tests and absence of recorded incidents each provide observations of quality, while leaving complete correctness unestablished. Validation must assess agreement between reviewers and document what the tests and rubric can detect.

For a finite intake cohort with positive total labour, define

$$P_H(\tau, \Delta) = \frac{\sum_i Q_i(\tau, \Delta)}{\sum_i H_i(\tau + \Delta)}. \tag{5}$$

The denominator is total cohort labour through the same task age $\tau+\Delta$ for every enrolment. It includes labour on rejected, unfinished and abandoned tasks, together with task-attributable work after acceptance. Early acceptance leaves more time for repair within this labour window than late acceptance. Stopping repair accounting exactly Δ after each acceptance would define a different denominator. State the chosen boundary and follow every enrolment for $\tau + \Delta$ to ascertain the numerator and denominator defined here.

Tasks with zero human labour can contribute output to the cohort ratio. The ratio is defined when aggregate labour is positive. Report its numerator and denominator separately.

3.3. **The observations and the three clocks.** The endpoints use three timing rules: acceptance is assessed against the deadline τ, retention at task age $T_i + \Delta$, and labour through task age $\tau + \Delta$. Calculating these endpoints from a complete history requires no assumption of Markov dynamics or independent attempts. The acceptance event alone leaves subsequent loss and repair unresolved. A realised labour total is point-identified by a record when every admissible history consistent with that record gives the same total. Proposition 2.1 instead concerns identification of a population parameter from the probability law of observations. We now derive the smallest and largest labour totals compatible with interval observations.

4. Attainable labour ranges from interval observations

An interval total leaves the timing of attention within it unresolved. We therefore seek all values of fixed-horizon labour compatible with the observations and the allocation constraint in (3). The *identified range* contains exactly those values: each is attained by at least one compatible history, and every compatible history gives a value in the range. This use of compatibility follows the general partial-identification perspective [12, Section 1]. The following proofs establish that range under the stated observation bounds and allocation constraints.

4.1. **Using the person's capacity.** Knowing which person supplied an interval total bounds how much can fit on either side of a cutoff. An aggregate across several people requires their combined capacity.

Proposition 4.1 (Attainable contribution from one interval)**.** *Let $\ell < u$, $c \in [\ell, u]$ and $v \in [0, u-\ell]$. Among measurable allocations $a : [\ell, u] \to [0,1]$ with $\int_\ell^u a(t)\,dt = v$, the attainable values of $x = \int_\ell^c a(t)\,dt$ are exactly*

$$[\max\{0, v-(u-c)\},\ \min\{v, c-\ell\}]\,. \tag{6}$$

If $v < 0$ or $v > u-\ell$, no such allocation exists.

Proof. Nonnegativity and capacity on the two pieces imply

$$0 \le x \le c-\ell, \qquad 0 \le v-x \le u-c.$$

Their intersection is (6). Conversely, for any x in this interval, set $a = x/(c-\ell)$ before c and $a = (v-x)/(u-c)$ after c. Each value lies in $[0,1]$ and gives the required integral. Omit a zero-length piece when c is an endpoint: the corresponding numerator is then zero. Finally, integrating $0 \le a \le 1$ gives the necessary condition $0 \le v \le u-\ell$. □

For example, 3/2 person-hours in a two-hour interval imply between 1/2 and 1 person-hours in its first hour. Nonnegativity alone would give $[0, 3/2]$. The capacity bound still leaves the allocation within the interval unresolved. If $v = u-\ell$, full utilisation identifies every cutoff integral from the interval total and capacity alone. For an interior subinterval $[b,c] \subseteq [\ell,u]$, the same construction on that subinterval and its complement gives the attainable marginal range $[\max\{0, v-((u-\ell)-(c-b))\}, \min\{v, c-b\}]$.

Marginal bounds can be summed when each contribution is counted once. The summed lower and upper bounds are attainable only if the respective component allocations can coexist. Disjoint intervals of one person, or intervals belonging to distinct people, can be combined when no other constraints link them. Overlapping observations or several tasks competing for the same person's time require a joint calculation. For example, two task totals of one person-hour each on $[0,2]$ individually allow $[0,1]$ before time 1. Their sum is exactly one person-hour before time 1: the shared capacity is saturated almost everywhere on $[0,2]$. Independently adding the marginal intervals would give $[0,2]$.

4.2. **An exact finite representation of compatible histories.** To express those shared constraints, call each task–person–role combination a stream. Fix finite sets of tasks $\mathcal{I}$, people $\mathcal{J}$ and roles $\mathcal{R}$, a cohort $\mathcal{C} \subseteq \mathcal{I}$, and a bounded calendar-time domain $[b_0, b_K]$ with $b_0 < b_K$ containing every observation interval and target window. An admissible history consists of measurable $a_{ijr} \ge 0$ satisfying $\sum_{i,r} a_{ijr}(t) \le 1$ almost everywhere for every person j. The declared roster and attribution rules define the labour being bounded, so evidence of their completeness is a measurement requirement. A residual category outside $\mathcal{C}$ can represent work that consumes capacity without contributing to the cohort target.

For observations $\nu = 1, \dots, M$, specify an interval $[\ell_\nu, u_\nu]$ within this domain, a set $G_\nu \subseteq \mathcal{I} \times \mathcal{J} \times \mathcal{R}$ of attributed streams, and finite labour bounds $v_\nu^- \le v_\nu^+$, in person-hours. Impose

$$v_\nu^- \le \sum_{(i,j,r)\in G_\nu} \int_{\ell_\nu}^{u_\nu} a_{ijr}(t)\,dt \le v_\nu^+. \tag{7}$$

Exact interval totals use $v_\nu^- = v_\nu^+$. Deterministic rounding or ascertainment-error limits may instead justify a nonzero width. A sampled Horvitz–Thompson estimate supplies an estimated total, whose sampling uncertainty requires a separate analysis before either inequality can be imposed.

Form the sorted distinct endpoints $b_0 < \cdots < b_K$ from the domain boundaries, every observation endpoint and every target boundary. For the cohort denominator, target boundaries include each cohort task's enrolment s_i and its fixed cutoff $c_i = s_i + \tau + \Delta$.

Translate task ages into calendar time before partitioning, so that simultaneous work by the same person shares one capacity constraint. Retain cells outside a target window whenever an observation spans them. Put $d_k = b_k - b_{k-1} > 0$ and let x_{ijrk} denote the labour allocated to stream (i, j, r) in cell k.

Proposition 4.2 (Exact cell reduction and attainable labour range)**.** *Under the preceding model, define the finite-dimensional feasible set $\mathcal{P}$ by*

$$x_{ijrk} \geq 0, \qquad \sum_{i,r} x_{ijrk} \leq d_k \qquad \text{for every } j, k, \tag{8}$$

$$v_\nu^- \leq \sum_{(i,j,r)\in G_\nu} \sum_{k:\,[b_{k-1},b_k]\subseteq[\ell_\nu,u_\nu]} x_{ijrk} \leq v_\nu^+ \qquad \text{for every } \nu. \tag{9}$$

Fix real weights w_{ijrk} and define the continuous target by weighting each cell integral by w_{ijrk}. For its finite representation $h(x) = \sum_{i,j,r,k} w_{ijrk} x_{ijrk}$, the attainable target values over admissible histories equal $h(\mathcal{P})$. In particular, the fixed-horizon cohort denominator uses $w_{ijrk} = 1$ exactly when $i \in \mathcal{C}$ and $[b_{k-1}, b_k] \subseteq [s_i, c_i]$, and zero otherwise. If $\mathcal{P}$ is nonempty, its target range is the closed interval

$$[H_{\min}, H_{\max}] = \left[\min_{x\in\mathcal{P}} h(x),\ \max_{x\in\mathcal{P}} h(x)\right]. \tag{10}$$

Both endpoints and every value between them are attained. If $\mathcal{P}$ is empty, the observations and capacity assumptions are inconsistent.

Proof. Integrating any admissible history over each cell gives nonnegative masses. Integrating the shared capacity gives (8). Splitting each observation at the cell boundaries gives (9). The target is preserved because its weights are constant on cells.

Conversely, for $x \in \mathcal{P}$ set $a_{ijr}(t) = x_{ijrk}/d_k$ in the interior of cell k, and set the finitely many endpoint values to zero. These measurable allocations are nonnegative and have sum at most one for each person. They reproduce every cell mass, observation and target value. Thus the histories and the polytope yield exactly the same attainable target values.

The constraints define a closed convex set. Each coordinate is between zero and d_k, so a nonempty $\mathcal{P}$ is compact and a linear functional attains its extrema. Convex combinations of minimising and maximising points attain every intermediate value. The reconstruction just given supplies an admissible history for each of them. □

The two extrema can be computed by linear programmes, each with $|\mathcal{I}||\mathcal{J}||\mathcal{R}|K$ variables, $|\mathcal{J}|K$ shared-capacity constraints and $2M$ observation inequalities, in addition to nonnegativity. Replacing the continuous history by cell totals preserves every attainable target value. The constant allocations used in the proof reconstruct compatible histories exactly. A known period of unavailability can be included by adding its boundaries to the partition and setting that person's cell masses to zero within it. This exactness depends on allowing measurable fractional allocations. Minimum session lengths or restrictions on task interruption would limit the reconstructed histories. Smoothness constraints and switching costs would also require a separate analysis.

For the cohort-denominator target, let $Q = \sum_{i\in\mathcal{C}} Q_i(\tau, \Delta)$ be exact retained output. With $Q \geq 0$ and $H_{\min} > 0$, monotonicity and attainability give the identified productivity interval $[Q/H_{\max}, Q/H_{\min}]$. Justified quality bounds $0 \leq Q_- \leq Q \leq Q_+$ give an outer interval $[Q_-/H_{\max}, Q_+/H_{\min}]$ containing every feasible ratio. It is the exact range only when every ratio in it is attainable from an admissible output–labour pair. Admissibility of the whole product $[Q_-, Q_+] \times [H_{\min}, H_{\max}]$ is sufficient. A zero lower labour bound does not justify dividing by it. The acceptance clock and retention assessment in (4) remain fixed throughout the allocation optimisation.

5. Estimating labour from sampled allocations

The preceding bounds treat interval totals or their limits as known. A probability sample instead supports an estimate whose properties concern repeated application of the sampling design. Substituting that estimate for an exact total can exclude the true allocation from the feasible set. The difference estimator below uses telemetry predictions while preserving design unbiasedness under explicit sampling and measurement assumptions.

Activity sampling estimates labour from selected person-intervals. The frame must cover the target work, including off-keyboard reasoning, review and coordination. Short diaries and task records can supply allocation observations, with independent assessment of a subsample to measure disagreement. Nonresponse, inaccurate recall and work outside the frame require separate investigation.

5.1. **Frozen predictions as auxiliary information.** The difference estimator starts with a fixed prediction of total labour and corrects it using a probability sample of allocation observations. Inverse inclusion probabilities make the weighted residual sum unbiased for total prediction error under the conditions below. Denis and Haddou give the fixed-prediction form and its design variance explicitly [6, Section 2.3, equations (2)–(4)]. Prediction-powered inference also identifies its mean construction as a difference estimator and considers finite-population extensions [1, Section 1.4 and Appendix B of the accompanying preprint].

Partition each person's target observation period into non-overlapping intervals. Let $U = \{1, \dots, M\}$, $M \geq 1$, be the complete, fixed frame of these person-intervals. For interval m, let $d_m > 0$ be its length in hours and $a_m \in [0, 1]$ its average allocation to the target work. Its labour is $y_m = d_m a_m$ person-hours, and the population total is $H = \sum_U y_m$. For every frame unit, retain a prediction f_m in person-hours. A raw SCH value requires a stated conversion to labour units before it can serve as f_m. The entire prediction vector must be fixed under the sampling design. Training on an independent earlier population can satisfy this condition: condition on the frozen predictions, then sample allocations to correct their total error.

Proposition 5.1 (Fixed-prediction difference estimator)**.** *For a probability sample $s \subseteq U$ with known first-order inclusion probabilities $\pi_m > 0$, set $r_m = y_m - f_m$ and*

$$\widehat{H}_{\mathrm{diff}} = \sum_{m \in U} f_m + \sum_{m \in s} \frac{y_m - f_m}{\pi_m}. \tag{11}$$

If y_m and f_m are fixed under the design and every sampled labour value is observed accurately, then $\mathbb{E}_s[\widehat{H}_{\mathrm{diff}}] = H$, where $\mathbb{E}_s$ averages over the sampling design. Neither independence between inclusion indicators nor correct predictions are required.

Proof. Write $I_m = \mathbf{1}\{m \in s\}$. Linearity and $\mathbb{E}_s[I_m] = \pi_m$ give

$$\mathbb{E}_s[\widehat{H}_{\mathrm{diff}}] = \sum_U f_m + \sum_U \frac{\mathbb{E}_s[I_m]}{\pi_m}(y_m - f_m) = \sum_U y_m.$$

□

The residual correction makes the estimate design-unbiased even when the fixed prediction total differs from H. Taking $f_m = 0$ gives the Horvitz–Thompson (HT) estimator [8],

$$\widehat{H}_{\mathrm{HT}} = \sum_{m \in s} \frac{d_m a_m}{\pi_m}. \tag{12}$$

Record sampled zero allocations as observations. An unanswered sampling prompt is nonresponse and must remain distinguishable from zero labour and from an unselected unit. The proof concerns true allocation values. Unbiasedness in the presence of measurement error requires additional assumptions on that error.

5.2. **Precision depends on residuals and the sampling design.** Let $\pi_{mk} = \mathbb{E}_s[I_m I_k]$, including $\pi_{mm} = \pi_m$. The exact design variance is

$$\operatorname{Var}_s(\widehat{H}_{\text{diff}}) = \sum_{m,k \in U} \frac{\pi_{mk} - \pi_m \pi_k}{\pi_m \pi_k} r_m r_k. \tag{13}$$

Expanding the variance of $\sum_U I_m r_m/\pi_m$ gives this expression. Unlike the unbiasedness proof, the variance calculation uses covariances between inclusion indicators, which require joint inclusion probabilities. The population identity in (13) also involves generally unknown unsampled residuals. Estimating it requires the actual sampling design and the joint-inclusion conditions of the chosen variance estimator.

For simple random sampling without replacement from $M \geq 2$ units, let the fixed sample size satisfy $1 \leq n \leq M$. Every size-n subset has the same probability and $\pi_m = n/M$. Write

$$S_r^2 = \frac{1}{M-1} \sum_U (r_m - \bar{r}_U)^2, \qquad S_y^2 = \frac{1}{M-1} \sum_U (y_m - \bar{y}_U)^2.$$

The inclusion covariances in (13) give

$$\operatorname{Var}_s(\widehat{H}_{\text{diff}}) = \frac{M^2}{n} \left(1 - \frac{n}{M}\right) S_r^2, \qquad \operatorname{Var}_s(\widehat{H}_{\text{HT}}) = \frac{M^2}{n} \left(1 - \frac{n}{M}\right) S_y^2. \tag{14}$$

For $2 \leq n \leq M$, an unbiased estimator of the design variance is

$$\widehat{V}_{\text{diff}} = \frac{M^2}{n} \left(1 - \frac{n}{M}\right) s_r^2, \qquad s_r^2 = \frac{1}{n-1} \sum_{m \in s} (r_m - \bar{r}_s)^2. \tag{15}$$

Indeed, $s_r^2 = [n(n-1)]^{-1} \sum_{m<k,\, m,k \in s} (r_m - r_k)^2$. Each pair has inclusion probability $n(n-1)/[M(M-1)]$, so $\mathbb{E}_s[s_r^2] = S_r^2$. A noncensus sample of size one does not supply this variance estimate. A complete census has zero sampling variance, including a single-unit population. Measurement error remains a separate question.

For $n < M$, the difference estimator improves on HT exactly when $S_r^2 < S_y^2$. The gain in sampling precision therefore depends on residual variation under the design. Consider $M = 4$, $n = 2$, and $y = (0, 2, 4, 6)$. The true total is 12, and HT variance is $80/3$. Frozen predictions $f = (1, 3, 3, 5)$ yield variance $16/3$, whereas $f = (6, 4, 2, 0)$ yield $320/3$. Enumerating all six samples reproduces these variances and unbiased totals. It also verifies that (15) averages to each corresponding variance. The second predictor increases variance fourfold. The difference estimator is design-unbiased with either predictor, while its precision differs. The finite checks establish the calculations for these two predictors. A workplace study must determine whether code-derived predictions reduce residual variation under its sampling design.

5.3. **Failure cases and implementation checks.** The fixed-vector proof does not apply when predictions are fitted to the realised sample. For $M = 2$, $n = 1$ and $y = (0, 2)$, suppose a fitted rule memorises the selected outcome and predicts zero for the other unit. Its observed residual is always zero. The difference total is nevertheless 0 or 2, each with probability $1/2$, giving expectation 1 rather than $H = 2$. A sample-trained method needs a separate argument under the actual finite-population design.

The estimate can also be negative even though true labour is nonnegative. With $y = (0, 0)$, $f = (10, 0)$ and one uniformly selected unit, (11) is -10 or 10 with equal probability. Clipping at zero changes its expectation from 0 to 5. Preserve the signed estimate when reporting this estimator and investigate an implausible result. Dividing retained output by an unbiased labour estimate also need not yield an unbiased productivity estimate.

The accompanying implementation preserves signed estimates and observed zeros, and records nonresponse separately from unselected units. It rejects incomplete declarations, nonresponse and inconsistent inputs. For a general declared design it reports point estimates. For simple random sampling it also reports variance and standard error under the

conditions above. These checks assess consistency of the supplied records. Establishing frame completeness, allocation accuracy and compliance with the declared design requires evidence from collection.

Exhaustive finite tests check the examples above, input validation and census boundaries. The implementation supplies no default confidence interval: unbiased variance estimation establishes neither a normal approximation nor finite-sample interval coverage.

6. Validating the measurements

An empirical study should first test whether the proposed observations can be collected accurately and with the required coverage. Before collection, register the task scope and attribution rules that determine which labour is counted. Fix the sampling frame, reference score, acceptance horizon and retention rubric for the same population. Measure allocation disagreement, nonresponse, follow-up completeness and collection burden by intervention, role and task class. Use these observations to choose sampling intensity and a precision target for a later intervention comparison.

For interval bounds, use independently assessed totals with justified ascertainment-error limits. At prespecified cutoffs, compare the marginal bounds with the joint range to measure how much shared capacity narrows the range. For sampled labour, compare design variance with and without predictions frozen before sampling. Preserve the inclusion probabilities and joint-design information needed for these uncertainty calculations.

6.1. **Comparing predictions with sampled labour.**

Sampling variation in a labour estimate contributes to its disagreement with a prediction. Compare frozen telemetry predictions by assigned intervention, role and task class on held-out teams or periods, while accounting for that sampling error. Correlation between binned averages leaves both systematic prediction error and individual prediction errors unresolved. The following identities show what a comparison against sampled labour can establish.

Treat each task's labour H in a finite validation population as fixed. Let f be its prediction, frozen before validation sampling, and suppose $\mathbb{E}_s[\widehat{H}] = H$ with finite design variance. Then

$$\mathbb{E}_s[f - \widehat{H}] = f - H,$$
$$\mathbb{E}_s[(f - \widehat{H})^2] = (f - H)^2 + \mathrm{Var}_s(\widehat{H}).$$

Expand the square: the cross term vanishes because $\mathbb{E}_s[\widehat{H} - H] = 0$. Dependence between tasks' samples does not alter these marginal identities, but affects aggregate uncertainty.

For a second prediction g frozen before the same validation sample, subtraction cancels the common sampling variance:

$$\mathbb{E}_s[(f - \widehat{H})^2 - (g - \widehat{H})^2] = (f - H)^2 - (g - H)^2.$$

Thus the observed squared-loss difference is design-unbiased for the true squared-loss difference. Estimating either loss level or the uncertainty of the difference requires further sampling-design information. Absolute error requires a separate analysis.

For example, eight one-hour person-intervals have allocations $(1, 1, 1, 1, 0, 0, 0, 0)$. Sample one uniformly, with inclusion probability $1/8$. True labour is $H = 4$, whereas its HT estimate is 0 or 8, each with probability $1/2$. A perfect prediction $f = 4$ has apparent absolute error 4 against the sampled estimate. The interval $[3.5, 4.5]$ covers true labour surely but covers the sampled estimate with probability zero. The signed residual remains unbiased for the true prediction error. Absolute error and interval coverage against $\widehat{H}$ describe agreement with the noisy sampled total.

To assess accuracy against true labour, use a denser, independently adjudicated validation subset and report its remaining measurement error, or state a measurement model that identifies the chosen accuracy measure. Retain inclusion probabilities and joint sampling-design information for uncertainty calculations.

Validation should also test the score's sensitivity to task splitting, squashing and representation. Report reviewer disagreement and follow-up completeness alongside prediction error. To assess the consequence of those uncertainties, vary plausible missing labour and disputed retention scores. If those variations reverse an intervention ranking, the intervention decision depends on resolving or accepting the measurement uncertainty.

6.2. **Conditions for an intervention comparison.** A subsequent causal study must define whether it concerns a fixed workload or an eligible portfolio whose intake can change under the intervention. Freeze task eligibility and the treatment of pre-assignment work for a fixed-workload comparison. Record postponed, rejected and abandoned requests. An organisation-wide adoption comparison instead needs a common calendar window and records of intake and work in progress. State the population and weighting for each contrast.

Shared reviewers, agent capacity and reusable components can transmit intervention effects between teams. The assignment design and analysis must account for those dependencies. Analyse tasks by assigned intervention and calculate uncertainty using the assignment units, with additional allowance for labour and quality subsampling. The number of commits supplies no additional independent assignments.

Use the same retention rubric and follow-up procedure across intervention groups. Duplicate assessment of a subset measures disagreement. Record failures of reviewer blinding and the timing of feedback, since assessment can trigger repair. Count that repair within the declared labour window. Extrapolating from an assessed subset requires known selection probabilities and assumptions addressing any effect of assessment on subsequent work. Report severe defects separately because a retained fraction cannot represent every security or reliability consequence.

Complete follow-up through $\tau + \Delta$ as required by Section 3. Distinguish permanent abandonment from loss of observation. If follow-up is incomplete, state the assumptions or bounds used to handle missing outcomes. Report labour, acceptance and retained output together, since reduced labour may accompany more abandonment. A productivity ratio needs its numerator, positive denominator and their joint uncertainty. Prespecify practical improvement thresholds and any quality noninferiority margin. Failure to detect a quality reduction does not establish that the margin is met.

7. Scope and next observations

The observation scheme determines which labour totals telemetry can identify. Commit events admit observationally equivalent activity laws. Interval totals or justified bounds, together with per-person capacity, define a set of compatible allocations. Partitioning at every observation endpoint and target cutoff preserves its complete range of labour totals in a finite optimisation problem. Summing marginal interval bounds can overlook restrictions imposed by shared capacity. Such a sum therefore gives a conservative range unless the component endpoints are jointly attainable.

Probability sampling permits design-unbiased estimation when the frame covers the target work, inclusion probabilities are known and sampled allocations are measured accurately. A fixed telemetry predictor improves precision only when its residuals reduce design variance. Diaries and task records can supply the observations, but their coverage of reasoning and coordination and the accuracy of their allocations require empirical evidence. The mathematical guarantees are consequently conditional on the measurement and sampling premises. Establishing a causal benefit from AI adoption requires a further comparison of interventions.

The next empirical test is whether task identities can connect all-role allocations to independently assessed retained output through the required follow-up. Allocation disagreement, missingness and collection burden will help determine how much measurement is practicable before an intervention trial is designed. That trial must also choose whether to assess a

fixed workload or an organisation-wide adoption that can change incoming requirements. The present results specify the labour bounds and sampling guarantees on which that study can build.

Reproducibility

The repository at https://github.com/emmett08/engineering-work-observatory/tree/main/paper contains the LaTeX source, exact rational interval checker and finite sampling checks. The analytical proofs establish the general results. The executable checks compare finite constructed cases with those results and do not validate workplace measurements. The application implements marginal labour bounds and the fixed-prediction estimator. It does not solve the joint shared-capacity programme. Its floating-point bounds have no outward-rounding guarantee, and its variance estimate is limited to the declared simple-random-sampling case. Input checks cannot establish frame completeness, allocation accuracy or compliance with a sampling design.

The unchanged historical package retains Lean proofs of the finite observation-law counterexample and inverse-probability-weighting algebra. The attainable interval bounds, joint cell reduction, difference-estimator variance and sampling-validation arguments developed here have the analytical proofs and computational checks described above. They lie outside the historical proof receipt's coverage.

Appendix A. A shortest-path solution for one allocation stream

For one scalar allocation $0 \le a \le 1$, the interval observations of Section 4 can be represented by cumulative labour at their endpoints. Using the sorted endpoints and cell lengths defined there, set $z_k = \int_{b_0}^{b_k} a(t)\,dt$. Write an inequality $z_q - z_p \le w$ as a directed edge $p \to q$ of weight w. Adjacent endpoints receive edges $k-1 \to k$ of weight d_k and $k \to k-1$ of weight zero. Observation bounds on $[b_p, b_q]$ add edges $p \to q$ of weight v^+ and $q \to p$ of weight $-v^-$. An exact mass v therefore gives edge weights v and $-v$. This is the standard difference-constraints representation [9, Slides 6–11].

Proposition A.1 (Attainable cumulative bounds from graph distances)**.** *The scalar observations admit an allocation if and only if this graph has no directed cycle of negative total weight. When it is feasible, let $D(p,q)$ be the shortest-path weight from p to q. For any endpoint pair $p \le q$, the identified range of $\int_{b_p}^{b_q} a(t)\,dt$ is*

$$[-D(q,p),\ D(p,q)]. \tag{16}$$

Proof. Summing edge inequalities around a cycle gives 0 at the left, so a negative cycle is inconsistent. Conversely, adjacent edges make every vertex reachable from every other. In the absence of negative cycles, all shortest-path weights are finite and $D(r,r) = 0$. For any root r, potentials $z_k = D(r,k)$ satisfy each edge inequality because $D(r,v) \le D(r,u) + w$ for an edge $u \to v$ of weight w. Subtracting z_0 normalises the initial cumulative value without changing differences. Adjacent-edge inequalities give increments in $[0, d_k]$. Constant allocations with these increments satisfy the observations.

Summing inequalities along paths yields $z_q - z_p \le D(p,q)$ and $z_q - z_p \ge -D(q,p)$. Potentials rooted at p attain the upper bound, and those rooted at q attain the lower bound. Convex combinations of their reconstructed allocations attain the intervening values. □

The reference checker uses exact rational arithmetic and an all-pairs shortest-path calculation. Given indexed, sorted endpoints, graph construction and this calculation use $O((K+1)^3 + M)$ arithmetic operations and $O((K+1)^2 + M)$ storage. Bit costs depend on the encoded rational inputs. The checker compares the bounds and the allocations attaining them against an independent enumeration of finite allocations, including overlapping interval observations and inconsistent inputs. These constructed examples check the calculation

against the proved result. Empirical measurements are still needed to validate the observation process. Several task streams sharing a person's capacity require the joint formulation in Proposition 4.2, because the scalar graph represents only one allocation.

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